

Appendix J: Assessed System Performance Reference

This study report partially meets NERC TPL-001-4 standard, NERC FAC-002-2 standard, and ATC Planning Criteria (PLG-CR-0001-V19.5). In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the interconnection who will sign GIA.

Table J.1 – NERC FAC-002-2 Compliance Table

Part	Item	Explanation	Evidence Location
R1.2	Reliability impact	This study determines the reliability impact of 18 MISO generator interconnection requests (J986, J1000, J1002, J1003, J1009, J1010, J1011, J1042, J1051, J1053, J1085, J1101, J1121, J1153, J1154, J1171, J1183, and J1188) on the ATC System.	Section 2.3 covers steady state thermal and voltage analysis results which include thermal and voltage constraints impacted by the new generating facilities. Section 3 and Section 6 covers dynamic/transient stability reliability impact of the new generating facilities Section 4 covers short circuit and breaker duty reliability impact of the new generating facilities.
	Applicable NERC Reliability Standards	The TPL-001-4 standard is selected as an applicable NERC Reliability Standard for this study. Portions of the reliability standard are applied when performing the system impact study for J986, J1000, J1002, J1003, J1009, J1010, J1011, J1042, J1051, J1053, J1085, J1101, J1121, J1153, J1154, J1171, J1183, and J1188.	NERC TPL-001-4 Reliability Standard, as described in Appendix A Section A.1 and Appendix J Table J.2.
	Regional planning criteria	This study adheres to study requirements outlined in the applicable version of MISO BPM 015 – Generator Interconnection. BPM 015 contains MISO Business Practices in implementing Attachment X of the MISO Tariff. In addition, PJM Performs System Impact Study using their methodology.	MISO Business Practice Manual, BPM-015-r21, as described in Appendix A Section A.1
	Transmission Owner planning standards	This study takes into account the applicable version of the ATC Planning Criteria and Assessment Practices.	ATC Transmission System Planning Criteria, PLG-CR-0001-V19.5, as described in Appendix A Section A.1 ATC Transmission Planning Assessment Practices, PLG-METH-0002-V7.3, as described in Appendix A Section A.1
R1.3	Facility interconnection requirements	This study also evaluates facility interconnection requirements such as voltage schedule, power factor, islanding detection and transformer energization in the applicable version of the ATC Generation Facility Interconnection Guide.	ATC Generating Facility Interconnection Guide, Revision 9.1, as described in Appendix A Section A.1
	Steady state analysis for normal and contingency conditions	Steady State Power flow analysis determines impacts of new generating facility on the line flow and bus voltages. This study identifies violations to line ratings and bus voltage limits in study models under normal and contingency conditions.	The steady-state contingencies that were evaluated for the SIS are described in Appendix A. Section 2.3 covers steady state thermal and voltage analysis results.
	Short circuit analysis for normal and contingency conditions	For existing substation and breakers, this study identifies areas where further investigation and potential upgrades and modifications will be required due new generator interconnections. It will be used to design new substation to be built for generator interconnections.	The short circuit conditions that were evaluated for the SIS are described in Appendix A. Section 4 covers short circuit and breaker duty analysis results.
	Dynamic analysis for normal and contingency conditions	This study simulates fault, relay and breaker actions near the generator and identifies any stability issues when new generators are interconnected to the grid. The results are used to develop mitigations and operating restrictions.	The dynamic stability contingencies that were evaluated for the SIS are described in Appendix A. Section 3 and Section 6 covers stability analysis results.
R1.4	Study assumptions	All relevant NERC Reliability Criteria and the ATC Planning Criteria are to be met for thermal, voltage and angular stability analysis.	Appendix A covers study criteria, methodology and assumptions. Section 2.1 covers modeling assumptions for the steady-state analysis. Section 3.1 covers modeling assumptions for the stability analysis. Appendix L covers modeling assumptions for the electromagnetic transient analysis.
	System performance	See R1.3	See R1.3
	Alternatives considered	The consideration of alternatives was applicable to ERIS and NRIS network upgrade development and select point of interconnection facilities.	Section 2.3.6 covers the alternatives that were considered for the ERIS thermal network upgrades. Section 6 covers the alternative that were considered for the ERIS EMT stability network upgrades. Section 8.3 covers the alternatives that were considered for the NRIS thermal network upgrades. Table J.3 covers the alternatives that were considered for point of interconnection facilities.
	Coordinated recommendations	ATC will submit a generator interconnection project with a signed GIA, along with any required network upgrades to MISO. Then MISO will publish the generator interconnection project, along with any required network upgrades, in its MTEP Appendix A, which represents MISO accepts the projects and authorizes its implementation.	To be completed in later DPP phases.

Table J.2 – MISO, ATC, NERC/FERC Studies Matrix

	MISO DPP						ATC Planning Criteria						NERC/FERC Compliance				
Study	PF	DYN	SC	BD	EMT	RAS	PF	DYN	SC	BD	EMT	RAS	PF	DYN ⁹	SC	BD	RAS
P0	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
P1	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
P2	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
P3	NO	NO	na	na	na	na	Yes ²	Yes ²	na	na	na	na	Yes ²	Yes ²	na	na	na
P4	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
P5	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
P6	NO	NO	na	na	na	na	Yes ¹	Yes	na	na	na	na	Yes ¹	Yes	na	na	na
P7	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na	na	Yes	Yes	na	na	na
E1	NO	NO	na	na	na	na	Yes ³	Yes ³	na	na	na	na	Yes ³	Yes ³	na	na	na
E2	NO	NO	na	na	na	na	Yes ³	Yes ³	na	na	na	na	Yes ³	Yes ³	na	na	na
E3	NO	NO	na	na	na	na	Yes ³	Yes ³	na	na	na	na	Yes ³	Yes ³	na	na	na
Power Factor Requirements	Yes	na	na	na	na	na	Yes	na	na	na	na	na	Yes	na	na	na	na
Transformer Energization	na	na	na	na	na	na	Yes	na	na	na	na	na	na	na	na	na	na
TPL-001-4 Requirement 4.3.1.3	NO	NO	na	na	na	na	Yes ²	Yes ²	na	na	na	na	Yes ²	Yes ²	na	na	na
Short Circuit	na	na	Yes	na	na	na	na	na	Yes	na	na	na	na	na	Yes	na	na
Breaker Duty	na	na		Yes	na	na	na	na	na	Yes	na	na	na	na		Yes	na
Models	SP&SH	SH ⁷	na	na	na	na	SP&SH	SP&LL ²	na	na	na	na	SP&SH	SP&LL ^{2&4}	na	na	na
Weak Grid Conditions	na	na	na	na	NO	na	na	na	na	na	Yes ⁵	na	na	na	na	na	na
SPS/RAS	NO	NO	na	na	na	NO	na	na	na	na	na	Yes ⁶	na	na	na	na	na
Power Quality Analysis	na	na	na	na	na	na	na	na	na	na	Yes ⁸	na	na	na	na	na	na
LVRT (Order 661/661-A)	na	Yes	na	na	na	na	na	Yes	na	na	na	na	na	Yes	na	na	na

¹ATC will perform NERC Category P6 steady-state thermal analysis to identify potential operating restrictions as part of DPP. ATC will perform NERC Category P6 steady-state voltage analysis outside of DPP process for the generators with signed the GIA in ATC Ten Year Assessment Process (TYA) annually.

²ATC will perform these studies outside of DPP process for the generators with signed the GIA in ATC TYA annually.

³ATC will rely on MISO annual MTEP Extreme Event study results for ATC TPL Extreme Event compliance. MISO MTEP study models include the generators with signed the GIA.

⁴TPL requires multiple models for dynamic analysis - ATC uses summer peak and light load models.

⁵ATC may perform electromagnetic transient (EMT) studies for select inverter-based resources in parallel with DPP studies.

⁶ATC will require assessment of existing SPS/RAS if there is potential for interaction with new interconnections.

⁷ATC will dispatch generators in the vicinity of new interconnections to Pmax.

⁸Developer is expected to provide power quality information to ATC for review. If deficient, studies will be required.

⁹Dynamic analysis may include EMT studies in selected situations.

Table J.3 – Point of Interconnection (POI) Alternatives Considered¹

MISO Queue #	POI Requested in Application	Alternative POI Considered	IC Final Decision on POI
J986	Port Edwards – Sand Lake 138-kV	Saratoga – Badger West 138-kV	Port Edwards - Sand Lake 138-kV
J1000	Nelson Dewey – Lancaster 138-kV	Nelson Dewey Substation was considered as a POI by both the interconnection customer and ATC. It was not selected due to the following difficulties of getting the 138-kV Gen-Tie line into Nelson Dewey substation from two potential project site areas: 1. The exiting Nelson Dewey-Lancaster-Hill Valley 138 kV will be rebuilt as 345/138 kV double circuits as part of the Cardinal – Hickory Creek project. Requesting an additional 138 kV circuit (J1000 Gen-Tie) on these structures is considered non-standard under ITC Midwest Design and Construction practices. 2. The existing Nelson Dewey – Potosi – Hillman 138 kV line structures cannot support a double circuit design. The line would need to be rebuilt at a higher than average cost due to the difficult terrain. The cost of re-building the existing line is expected to be higher than building a new 138 kV switching station for the proposed POI.	Nelson Dewey - Lancaster 138-kV
J1002	Wautoma 138-kV	Wautoma – Sand Lake 138-kV	Wautoma 138-kV
J1003	North Beaver Dam 69-kV	New substation on one of the lines out of North Beaver Dam	North Beaver Dam 69-kV
J1009	Pleasant Prairie-Racine 345-kV	Mount Pleasant 138 kV	Pleasant Prairie-Racine 345-kV
J1010	Arcadian-Zion 345-kV	N/A	Arcadian-Zion 345-kV
J1011	Arcadian-Zion 345-kV	N/A	Arcadian-Zion 345-kV
J1042	North Lake Geneva 138-kV	N/A	North Lake Geneva 138-kV
J1051	Concord 138-kV	New substation on one of the lines out of Concord	Concord 138-kV
J1053	Concord 138-kV	New substation on one of the lines out of Concord	Concord 138-kV
J1085	Stone Lake – Gardner Park 345 kV line, 26 miles from Gardner Park	Combined POI with J1121	Gardner Park-Stone Lake 345-kV
J1101	Kewaunee 138-kV	N/A	Kewaunee 138-kV
J1121	Stone Lake – Gardner Park 345 kV line, 20 miles from Gardner Park	Stone Lake – Gardner Park 345 kV line, 26 miles from Gardner Park. Combined POI with J1085	Gardner Park-Stone Lake 345-kV
J1153	Holland 138-kV	N/A	Holland 138-kV
J1154	Jefferson 138-kV	N/A	Jefferson 138-kV
J1171	Butternut 138-kV	N/A	Butternut 138-kV

J1183	Garden Corners 138-kV	N/A	Garden Corners 138-kV
J1188	Sheepskin 69-kV	N/A	Sheepskin 69-kV

¹ Based on MISO Tariff, when filing application to the interconnection queue, Interconnection customer (IC) can select any Point of Interconnection (POI) at a location under MISO's functional control. This could be an existing substation or a new substation tapping the existing line. Further in the process, IC could unilaterally change POI to a new location within 5 days after the scoping call. Scoping calls occur within 45 days prior the start of Definitive Planning Phase (DPP). Any change in POI past the DPP start date requires consent for all parties i.e. MISO, TO, and IC.

Prior to the scoping calls, ATC review IC's POI selection, initiates a conversation with IC and MISO and provides alternative POI options for consideration where warranted. ATC proposes alternative POI options for variety of reasons including, but not limited to, where the site constraints will make it difficult to expand an existing substation or build a new one, expanding the existing substation maybe be high cost compared to building a new site, connecting to a nearby existing substation maybe be an overall cheaper option compared to building a new site, or where combing several ICs relatively close to each other to a single site would be cost effective.

Ultimately, after the initial conversations, IC decides to pursue those option further or not. ATC offers help if further assistance is needed to evaluate various POI alternatives.